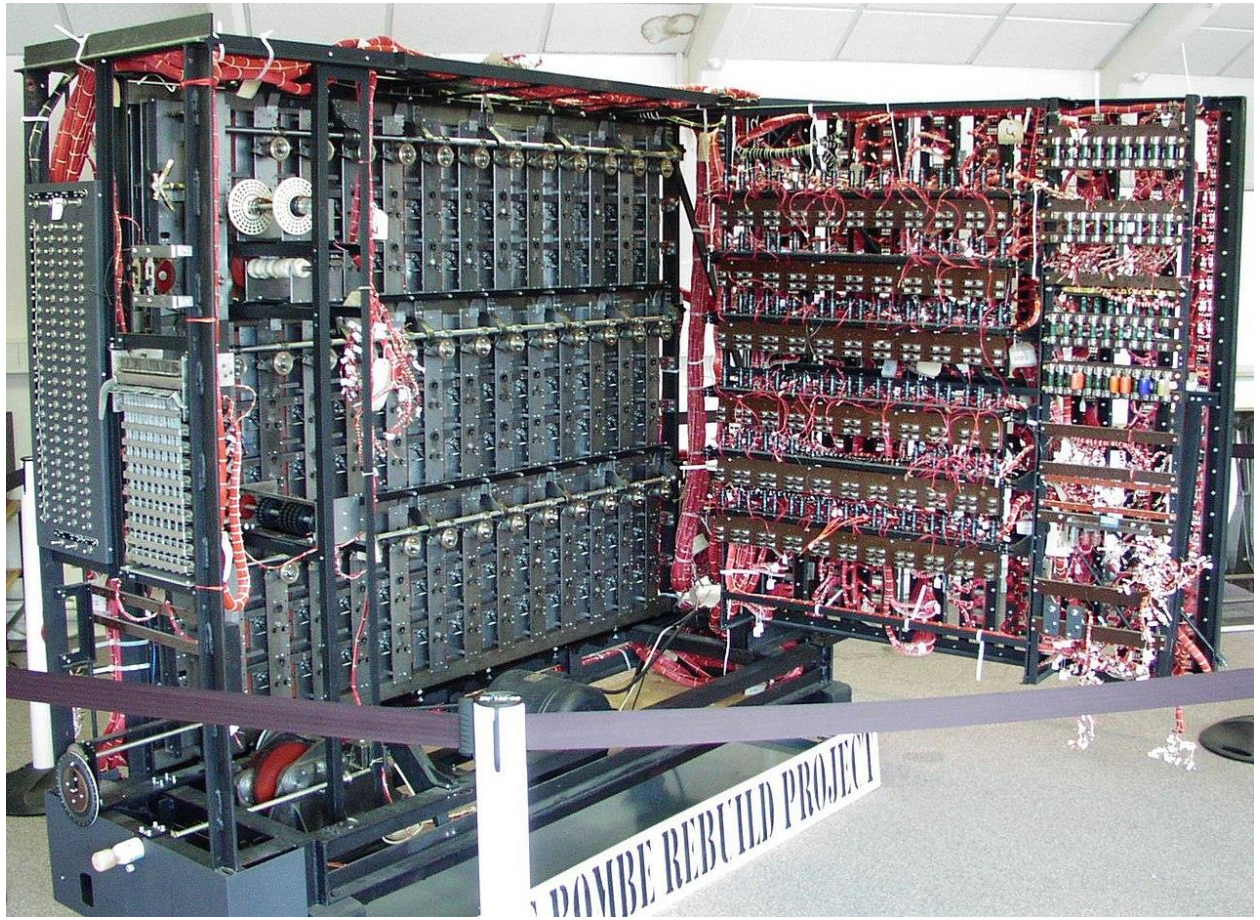
ПЛАМЕН СПАХИЕВ

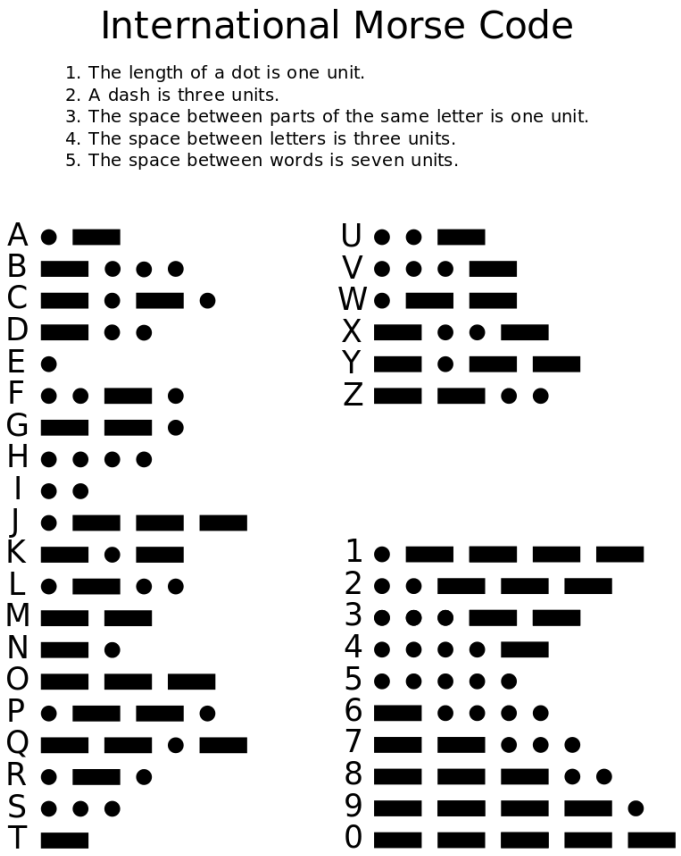
ФПМИ

БРОЙНИ СИСТЕМИ И КОНВЕРТИРАНЕ

ПРИЛОЖЕНИЕ НА БРОЙНИТЕ СИСТЕМИ В КОМПЮТРИТЕ



ПРИЛОЖЕНИЕ НА БРОЙНИТЕ СИСТЕМИ В КОМПЮТРИТЕ



Фиг. 1. Морзовата азбука.

ПРИЛОЖЕНИЕ НА БРОЙНИТЕ СИСТЕМИ В КОМПЮТРИТЕ

USASCII code chart

b_7 b_6 b_5 →					0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
Bits	b_4	b_3	b_2	b_1	Column → / Row ↓	0	1	2	3	4	5	6	7
	0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
	0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
	0	0	1	0	2	STX	DC2	"	2	B	R	b	r
	0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
	0	1	0	0	4	EOT	DC4	$	4	D	T	d	t
	0	1	0	1	5	ENQ	NAK	%	5	E	U	e	u
	0	1	1	0	6	ACK	SYN	&	6	F	V	f	v
	0	1	1	1	7	BEL	ETB	'	7	G	W	g	w
	1	0	0	0	8	BS	CAN	(	8	H	X	h	x
	1	0	0	1	9	HT	EM	)	9	I	Y	i	y
	1	0	1	0	10	LF	SUB	*	:	J	Z	j	z
	1	0	1	1	11	VT	ESC	+	;	K	[	k	{
	1	1	0	0	12	FF	FS	,	<	L	\	l	|
	1	1	0	1	13	CR	GS	–	=	M	]	m	}
	1	1	1	0	14	SO	RS	.	>	N	^	n	~
	1	1	1	1	15	SI	US	/	?	O	_	o	DEL

Фиг. 2. ASCII таблица преди 1972 година

ДВОИЧНА БРОЙНА СИСТЕМА

0 1 2 3 4 5 6 7 8 9 10 11 12 13……19 20 21 22 23…….98 99 100 101 102 103……

10^4 10^3 10^2 10^1 10^0

$23475 = 2x10^4 + 3x10^3 + 4x10^2 + 7x10^1 + 5x10^0$

$10^0 = 1$

$10^1 = 10x1 = 10$

$10^2 = 10x10 = 100$

$10^3 = 10x10x10 = 1000$

$10^4 = 10x10x10x10 = 10000$

$10^5 = 10x10x10x10x10 = 100000$

$10^6 = 10x10x10x10x10x10 = 1000000$

$10^7 = 10x10x10x10x10x10x10 = 10000000$

$10^8 = 10x10x10x10x10x10x10x10 = 100000000$

$10^9 = 10x10x10x10x10x10x10x10x10 = 1000000000$

ДВОИЧНА БРОЙНА СИСТЕМА

1 10 11 100 101 110 111 1000 1001 1010 1011 1100 1101 1111

$11001 = 1x2^5 + 1x2^4 + 0x2^2 + 0x2^1 + 1x2^0$

$2^0 = 1$

$2^1 = 2x1 = 2$

$2^2 = 2x2 = 4$

$2^3 = 2x2x2 = 8$

$2^4 = 2x2x2x2 = 16$

$2^5 = 2x2x2x2x2 = 32$

$2^6 = 2x2x2x2x2x2 = 64$

$2^7 = 2x2x2x2x2x2x2 = 128$

$2^8 = 2x2x2x2x2x2x2x2 = 256$

$2^9 = 2x2x2x2x2x2x2x2x2 = 512$

1	=	1
2	=	10
3	=	11
4	=	100
5	=	101
6	=	110
7	=	111
8	=	1000
9	=	1001
10	=	1010
11	=	1011
12	=	1100
13	=	1101

ШЕСТНАДЕСЕТИЧНА БРОЙНА СИСТЕМА

0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F
16	10

Табл. 1. Паралел между десетична и шестнадесетична система

$16^0 = 1$

$16^1 = 16x1 = 16$

$16^2 = 16x16 = 256$

$16^3 = 16x16x16 = 4096$

$16^4 = 16x16x16x16 = 65536$

$16^5 = 16x16x16x16x16 = 1048576$

$16^6 = 16x16x16x16x16x16 = 16777216$

$16^7 = 16x16x16x16x16x16x16 = 268435456$

$16^8 = 16x16x16x16x16x16x16x16 = 4294967296$

(ffffffff = 4294967296)

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ДВОИЧНА СИСТЕМА

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

10110111

128	64	32	16	8	4	2	1
1	0	1	1	0	1	1	1

(1x128) + (0x64) + (1x32) + (1x16) + (0x8) + (1x4) + (1x2) + (1x1) = 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 = 183

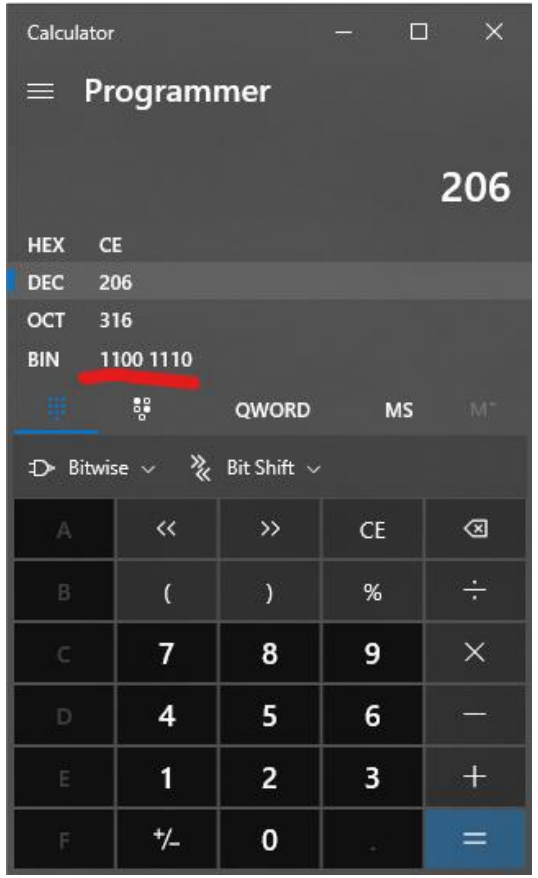
10110111 = 128

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ДВОИЧНА СИСТЕМА

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	1	0	0	1	1	1	0

128+64+8+4+2=206

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ДВОИЧНА СИСТЕМА



Фиг. 3. Конвертиране в двоична система чрез Calculator на MS Windows 10.

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ШЕСТНАДЕСЕТИЧНА СИСТЕМА

16^4	16^3	16^2	16^1	16^0
65536	4096	256	16	1
2	3	c	a	5

(2x65536) + (3x4096) + (12x256) + (10x16) + (1x5) = 131072 + 12288 + 3072 + 160 + 5 = 146597

23ca5 = 146597

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ШЕСТНАДЕСЕТИЧНА СИСТЕМА

7554

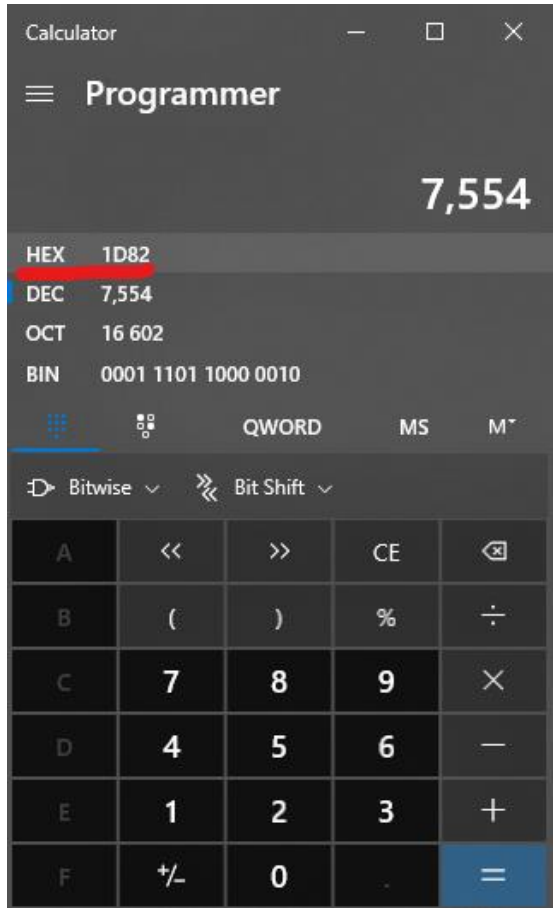
16^4	16^3	16^2	16^1	16^0
65536	4096	256	16	1
	1	d	8	2

(1x4096) + (13x256) + (8x16) + (1x2) = 4096+3328+128+2 = 7554

7554 = 1d82

0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F
16	10

КОНВЕРТИРАНЕ НА ДЕСЕТИЧНА В ШЕСТНАДЕСЕТИЧНА СИСТЕМА



Фиг. 4. Шестнадесетично представяне на числото 7554

КОНВЕРТИРАНЕ НА ДВОИЧНА В ШЕСТНАДЕСЕТИЧНА СИСТЕМА

11011011

1101 1011

8 4 2 1

1 1 0 1

8+4+1=13=d

8 4 2 1

1 0 1 1

8+2+1=11=b

11011011=128+64+0+16+8+0+2+1=219

db=(16x13)+(1x11)=208+11=219

219=219 => 11011011=db

MAC АДРЕСИ

- Organizationally Unique Identifier **(OUI)** - производителя на устройството
- Universally Administered Address **(UAA)** - информация за самото устройство

MAC48
EUI64 -> IPv6

Фиг. 5. Етикет на рутер, даващ информация за неговите LAN и WLAN MAC адреси.

MAC АДРЕСИ

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wireless-AC 9560 160MHz
   Physical Address. . . . . . . . . : 60-F2-62-81-7B-67
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::f56f:7fdc:9bb4:a5c2%24(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.100.100.150(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, October 17, 2021 18:20:31
   Lease Expires . . . . . . . . . . : Sunday, October 17, 2021 21:30:52
   Default Gateway . . . . . . . . . : 10.100.100.1
   DHCP Server . . . . . . . . . . . : 10.100.100.1
   DHCPv6 IAID . . . . . . . . . . . : 207680098
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-25-DD-F3-FB-F8-75-A4-9F-86-EC
   DNS Servers . . . . . . . . . . . : 10.100.100.1
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Фиг. 6. Резултат от ipconfig /all под Windows

MAC АДРЕСИ

```
plamen@UirtualBox:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:68:cc:62 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86131sec preferred_lft 86131sec
    inet6 fe80::6e85:bcda:d81d:b199/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:2b:a1:77:45 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
plamen@UirtualBox:~$
```

Фиг. 7. Резултат от ip address под Linux